Cross-Traffic Management System

System Requirements and Use Case Diagram

Team B6

System Requirements

Functional Requirements

FR-01 — The system shall detect vehicles approaching the intersection from all four directions.

FR-02 — The system shall model vehicle queuing using an M/M/c queuing system, where c is a configurable parameter representing the number of simultaneously permitted non-conflicting vehicles. The service discipline shall be a design parameter of the queuing model.

FR-03 — The system shall grant or deny intersection access to each vehicle based on the current queue state and conflict analysis.

FR-04 — The system shall behave deterministically; identical system states and inputs shall always produce identical outputs and decisions.

FR-05 — The system shall respond to all vehicle arrival, queuing, and crossing events within a defined and bounded time interval.

FR-06 — The system shall detect conflicting vehicle paths before granting access and shall not permit two vehicles whose paths intersect to cross simultaneously.

FR-07 — The system shall continuously monitor queue length and trigger congestion mitigation when a defined threshold is exceeded.

FR-08 — The system shall formally register each vehicle exit from the intersection zone and update the queue state accordingly.

FR-09 — The system shall support Vehicle-to-Vehicle (V2V) communication to allow vehicles to broadcast their position and state to other nearby vehicles.

FR-10 — The system shall support Vehicle-to-Infrastructure (V2I) communication to allow vehicles to exchange signals with roadside units.

FR-11 — The system shall evaluate and compare the performance of the queuing system under different values of c to determine the optimal configuration for congestion reduction.

Non-Functional Requirements

NFR-01 — The system shall operate within the resource constraints of an embedded hardware platform including memory, processing power, and energy.

NFR-02 — The system shall scale to handle a minimum of four simultaneous vehicle queues, one per approach direction.

NFR-03 — The system implementation shall maintain a well-defined and traceable mapping to the formal model, regardless of the implementation language used.

NFR-04 — The system shall maintain safe and correct operation under peak vehicle arrival rates without performance degradation or safety violations.

NFR-05 — The V2V and V2I communication channels shall be treated as reliable and lossless within the simulation scope.

Use Case Diagram